
Statistical Inference

B. Statistical Data Science 2nd Year Indian Statistical Institute

End-semester Examination

15 December 2025

10:30 am – 1:30 pm

Name:

Roll Number:

Instructions

- Answer as much as you can. The maximum you can score is 50.
 - You can use any result obtained in the class. Clearly state the results that you use.
 - Clearly state all the assumptions that you make. Failure to do so will lead to deduction of marks.
 - Keep your answers to the point. Any unnecessary or irrelevant writing will be penalized.
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Question 1.

Total marks: 15

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ are unknown.

- (a) Derive the uniformly minimum variance unbiased estimator (UMVUE) of σ^2 . Also derive the variance of the UMVUE. [3+3 marks]
- (b) Find the maximum likelihood estimator (MLE) of σ^2 . Derive the bias and the variance of the MLE. [2+2+3 marks]
- (c) Among the UMVUE and the MLE, which one has a lower mean-squared error? [2 marks]

Question 2.

Total marks: 15

Let $X_1, \dots, X_n$ be independent random variables with $X_i \sim \text{Uniform}(0, i\theta)$, $i = 1, \dots, n$.

- (a) Identify a statistic $T = T(\mathbf{X})$ such that the joint distribution of $X_1, \dots, X_n$ has monotone likelihood ratio (MLR) property with respect to T . [3 marks]
- (b) Using part (a), or otherwise, construct a uniformly most powerful (UMP) size α test for testing $\mathcal{H}_0 : \theta \geq 1$ vs. $\mathcal{H}_a : \theta < 1$. You need to specify the exact value of the cut-off. [5 marks]
- (c) Construct a $100(1 - \alpha)\%$ confidence interval for θ of the form $[a, \infty)$. [7 marks]

Question 3.Total marks: 12

We are interested in comparing the production lines of two bulb producing companies: Company A and Company B. Based on a pilot survey, we have gathered the following information about the bulbs produced by the two companies.

	Company A	Company B
Total no. of bulbs produced	N_A	N_B
No. of fused bulbs	F_A	F_B

- (a) Construct a test for testing whether the proportions of fused bulbs produced by the two companies are the same or not. Clearly define the test statistic and the rejection rule. Justify your choice of test. [6 marks]
- (b) Construct a confidence interval for the difference of the proportions of fused bulbs produced by the two companies. Give justification for your answer. [6 marks]

Question 4.Total marks: 18

The blood groups of human beings are determined by the presence/absence of two different antigens, A and B. The following chart shows the determination of the four blood groups, A, B, AB and O, depending on the antigens.

	Antigen A present	Antigen A absent
Antigen B present	Blood group AB	Blood group B
Antigen B absent	Blood group A	Blood group O

Let p_A, p_B, p_{AB} and p_O denote the probabilities of a (randomly selected) person having blood group A, B, AB, or O, respectively. We select 500 people at random and note down their blood groups.

- (a) Formulate a statistical model for this data. Find MLEs of p_A, p_B, p_{AB} and p_O . [3+5 marks]
- (b) We want to test whether the probabilities of observing one of the four blood groups are the same. Formulate this as a statistical hypothesis testing problem. Derive a test for this problem. Clearly specify the test statistic and the rejection rule. Justify the choice of your test. [2+8 marks]

Hint: $p_A + p_B + p_{AB} + p_O = 1$.